

SilentMethyl: Multimodal Prediction of CpG Methylation and Single-Nucleotide Variant–Associated Methylation Changes, with Calibrated Uncertainty and Independent Cross-Tissue Variant Validation

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Abstract

DNA methylation is shaped both by local sequence and by the chromatin context in which that sequence sits, but establishing which single-nucleotide variants actually change methylation remains difficult. We present SilentMethyl, a gated fusion of a pretrained DNA language model with reference epigenomic features that predicts CpG methylation and the methylation change a variant causes. Under chromosome-blocked splitting the model reaches a β -value MAE of 0.0993 and ROC-AUC of 0.968 on 26,570 held-out CpGs, outperforming six comparators including the CpGenie and DeepCpG architectures retrained on identical data. Predicted variant effects replicate across two independent cohorts — 66,495 variant–CpG pairs from an African American whole-blood meQTL study and 76,893 from tissue-matched breast — agreeing in direction and rank, surviving matching on variant–CpG distance, and pooling with no detectable heterogeneity. Two findings generalise beyond this model. Gated fusion improves absolute prediction but contributes nothing to variant-effect prediction, because context features are identical for both alleles and therefore cannot create a variant effect; and uncertainty on bounded bimodal targets is confounded by range compression, so calibration must be assessed in logit space. Predicted effect magnitude does not by itself identify clinically annotated variants.

1 Introduction

DNA methylation is a widely studied epigenetic modification involved in gene regulation, shown to be associated with genomic context, chromatin organization, and transcription [1, 2], and frequently dysregulated in disease. Single-nucleotide variants (SNVs) can influence local epigenetic states, but validating their direct regulatory effects is difficult because matched genetic and epigenetic data are limited. This motivates modeling DNA methylation to identify potentially functional variants.

Existing deep learning methods have been able to predict DNA methylation and other similar regulatory measures using local DNA sequence, neighboring methylation, and long-range sequence context[3–5]. While DNA sequence itself contains some motifs and local regulatory patterns, sequence

alone misses the epigenetic context in which they operate. Reference epigenomic data, including chromatin accessibility, histone data, and evolutionary conservation, can account for some of this missing information. Predicting methylation at a single locus and estimating the effect of a base change on methylation are also distinct tasks. A model that accurately predicts baseline methylation may still estimate variant-induced changes poorly.

To address this, several frameworks for evaluating paired sequence changes already exist. CpGenie predicts CpG methylation and validates REF-ALT sequence changes against mQTLs[6]. INTERACT focuses on methylation in brain tissue, DeepMethylation/DDM combines DNA sequence with reference tissue data, and Metheven combines sequence with ATAC-seq chromatin accessibility[7–9].

Long-context models such as Enformer and AlphaGenome predict many regulatory modalities but not CpG methylation directly [5, 10], whereas MethylGPT operates on measured methylation profiles and cannot score an unobserved substitution [11]. Sequence-based methylation-effect prediction therefore remains a distinct problem.

However, many of these existing approaches face important roadblocks. Existing studies generally quantify the contribution of sequence and epigenomic inputs only through aggregate model-performance differences. Previous works also fail to account for reverse-complement equivariance, leaving them blind to this important biological phenomenon[12]. SilentMethyl was designed to address these gaps. Our framework uses gated fusion to show the relative importance of sequence and context information for individual probes. The model also enforces reverse complement stability through random orientation during training.

SilentMethyl also uses synonymous mutations as a focused baseline for variant prioritization, since they modify a base pair without changing the encoded amino acid. However, sSNVs can still affect local regulatory function, including splice recognition, RNA structure, transcript stability, translation, and regulatory sequences in exons. [13, 14]. Reported synonymous variants in *MLH1*, *TP53*, and *ADAMTS13* have been

linked to altered methylation or CpG content [15–17], motivating their use as focused targets for model experimentation.

Because prior studies differ in tissue, targets, sequence windows, and data splits, their metrics are not directly comparable. SilentMethyl instead uses chromosome-held-out data to train a gated sequence–context model, validates paired WT–MUT predictions against external mQTLs, and ranks variants with potential local methylation effects.

The expanded evaluation tests 66,495 held-out GENOA pairs alongside tissue-matched eGTEx data and uses distance matching, block resampling, published-model comparisons, and motif and clinical controls to define the limits of variant-response ranking.

2 Methods

2.1 Data construction

Methylation beta labels were constructed from the TCGA-BRCA HM450 matrix [18, 19]. Using a sample-type-11 filter, 97 solid-tissue-normal columns were selected from 893 total, verified to represent 97 unique participants without duplicate samples.

Single-probe methylation was represented as the median of patient beta values. For CpG i , if O_i is the set of patients with a recorded measurement at CpG i , then

$$\beta_i = \text{median}_{s \in O_i} \beta_{i,s}, \quad y_i = \mathbf{1}[\beta_i > 0.5],$$

$$\tilde{\beta}_i = \max(10^{-4}, \min(1 - 10^{-4}, \beta_i)),$$

$$M_i = \log_2 \left(\frac{\tilde{\beta}_i}{1 - \tilde{\beta}_i} \right).$$

The beta-to- M transformation follows [20].

All probes required an exact CpG-centered DNA sequence and a valid reference context window. None of the 418,486 probes in the combined dataset were flagged by the HM450 MASK_general filter. Median probe coverage was 97 of 97, and 95.1% of loci had at least 78 valid patients. Probe coordinates and sequence were taken from the hg38 HM450 annotation resource [21, 22].

Data were split by chromosome before training: chromosomes 10–11 for validation, chromosomes 8–9 for test, remaining chromosomes for training (Supplementary Table 1). Splitting by chromosome ensures no sequence windows overlap between partitions.

Each DNA input was a 1,000-bp CpG-centered sequence with the target CpG nucleotides at zero-indexed positions 499 and 500. The epigenetic input comprised seven MCF-10A attributes—ATAC-seq accessibility and six histone marks (H3K4me3, H3K27ac, H3K27me3, H3K9me3, H3K36me3, H3K4me1)—averaged across a CpG-centered 100-bp window, plus two PhyloP scores for the two nucleotides of the central CpG. A missingness mask accompanied each feature; missing values were imputed with the feature’s global median.

Reference data were derived from shared resources and were not patient-specific.

2.2 Models, training, and evaluation

Figure 1 summarises target construction, the chromosome partition, the gated architecture, and the paired scoring and evaluation paths described in this section. The sequence module comprised a pretrained DNABERT-2 (117M) transformer [23] followed by a convolutional layer and masked attention pooling:

$$\alpha_t = \frac{\exp(s_t)}{\sum_{u \in V} \exp(s_u)}, \quad \mathbf{h}_{\text{seq}} = \sum_{t \in V} \alpha_t \mathbf{e}_t,$$

where V contains non-padding positions and $\mathbf{e}_t$ is the embedding at position t , producing a 768-dimensional $\mathbf{h}_{\text{seq}}$.

The context module was an MLP mapping the 18-dimensional epigenetic input (9 features + 9 missingness masks) to a 768-dimensional $\mathbf{h}_{\text{ctx}}$.

After LayerNorm produced $\tilde{\mathbf{h}}_{\text{seq}}$ and $\tilde{\mathbf{h}}_{\text{ctx}}$, the two were concatenated and passed through a gating MLP yielding scalars g_{seq} and g_{ctx} . The fused embedding was

$$\mathbf{z} = g_{\text{seq}} \tilde{\mathbf{h}}_{\text{seq}} + g_{\text{ctx}} \tilde{\mathbf{h}}_{\text{ctx}},$$

fed to two prediction heads. The regression head predicted M , converted by $\hat{\beta} = 2^{\hat{M}} / (1 + 2^{\hat{M}})$; the classification head predicted $\hat{y} = \mathbf{1}[\sigma(\text{logit}) > 0.5]$. The objective was Huber loss ($\delta = 1.345$) on regression plus binary cross-entropy on classification.

Sequence and context models were trained independently with AdamW, with DNABERT-2 fully fine-tuned. Sequence, context, and fusion training used up to 10, 15, and 10 epochs at learning rates 2×10^{-5} , 10^{-3} , and 5×10^{-4} . During fusion training the component encoders were frozen and only normalization, gating, and prediction modules were updated.

During training, forward or reverse-complement orientation was chosen with equal probability per probe. For seeds 42, 43, and 44 the checkpoint was selected by lowest validation beta MAE. Primary metrics were RMSE and MAE for β and M , and ROC-AUC for classification.

The trained models were then used for paired sequence scoring. For each SNV, the predicted methylation difference was averaged across strands:

$$\Delta\hat{\beta} = \frac{(\hat{\beta}_{\text{MUT,FWD}} - \hat{\beta}_{\text{WT,FWD}}) + (\hat{\beta}_{\text{MUT,RC}} - \hat{\beta}_{\text{WT,RC}})}{2},$$

with $\Delta\hat{M}$ defined analogously. Because the reported prediction is $\frac{1}{2}[f(x) + f(\text{RC}(x))]$, it is exactly invariant under reverse complementation.

Tagged somatic sSNVs were obtained through the GDC with amino-acid preservation verified against hg38 and GENCODE v44 [19, 24]. Variants were matched to the nearest HM450 probe; 440 unique variants remained after removing MASK_general-flagged probes and variants outside the 1,000-bp window. Variants were prioritized by $|\Delta\hat{\beta}|$ from the three-seed fusion ensemble.

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Figure 1: SilentMethyl data construction, model architecture, and evaluation workflow. Panel A shows methylation target construction, chromosome partitioning, and model inputs. Panel B shows FWD-RC orientation and staged training. Panel C shows the gated multimodal architecture. Panel D shows paired WT-MUT scoring, held-out probe testing, external mQTL validation, and synonymous-variant ranking.

After model evaluation, we grouped the held-out CpGs by biological context. Each CpG was assigned to one GENCODE region category in the following order: promoter/TSS, UTR, gene body, or intergenic. CpGs were also grouped by their relation to CpG islands and by ATAC and H3K27ac signal quartiles. Within each group, we measured the benefit of reference context as the sequence-only beta MAE minus the fusion beta MAE, so a positive value indicated better fusion performance. Confidence intervals were estimated using 2,000 resamples of complete 1-Mb genomic regions. Candidate variants were separately grouped by genomic region and distance from the paired CpG. These groupings were used only to interpret model behavior and did not affect training, model selection, or candidate ranking.

2.3 External variant evaluation and statistical analysis

External validation used GENOA cis-meQTL summary statistics from an African American whole-blood cohort [25]. SNP positions were converted from hg19 to hg38, 0.1% of allele-inconsistent pairs were discarded, and effects reported per minor allele [26] were re-signed to the model’s REF→ALT direction. The independent held-out stratum contained 66,495 pairs (39,657 variants and 19,081 probes). Variants outside the window or at the target CpG were excluded; primary results use the 42,866 pairs that neither create nor destroy a CpG.

Confidence intervals resampled complete 1-Mb blocks to preserve local dependence. Because significant GENOA pairs were closer to their CpGs than null pairs (median 191 versus 258 bp), each significant pair was matched without replacement to a null at the same distance. Distance therefore had AUROC 0.500 in the matched cohort. Model differences were evaluated on one shared matched cohort with paired block bootstrap intervals.

The complete eGTEX association set provided a secondary cohort-wide analysis. After allele alignment and sequence-window filtering, 76,893 held-out variant-CpG pairs

remained; 418 significant non-CpG-altering pairs met the study’s two-stage permutation threshold and formed the primary tissue-matched stratum. Signed correlation, direction agreement, calibration slope, and AUROC were evaluated with 1-Mb block resampling.

Significant eGTEX Breast Mammary Tissue lead mQTLs ($q < 0.05$) were intersected with CpGs on the held-out test chromosomes [28]. Variants outside the 1,000-bp sequence window or altering the target CpG were excluded, leaving 81 associations from 70 unique variants. Predicted ALT-minus-REF methylation changes were compared with reported ALT-dosage slopes. Signed Spearman correlation measured agreement in the relative ordering of signed responses, sign concordance measured the fraction of predictions with the correct direction, and AUROC measured separation of positive and negative slopes. Absolute Spearman correlation assessed whether predicted and observed response magnitudes had similar ordering. A probe-overlap sensitivity analysis additionally excluded variants within the annotated rHM450 probe span or at either immediately adjacent nucleotide.

We also tested whether predicted response magnitude distinguished significant mQTLs from nonsignificant high- q leads ($q \geq 0.5$). Significant and high- q variants were matched one-to-one on the same chromosome, with an MAF difference no greater than 0.05 and preference for similar variant-CpG distance, baseline methylation, and substitution type. The procedure produced 35 pairs from pools of 53 significant and 51 high- q variants. AUROC and average precision tested whether $|\Delta\hat{M}|$ or $|\Delta\hat{\beta}|$ separated the two groups. Pair resampling produced confidence intervals, and within-pair label permutations produced p -values.

2.4 Additional analyses and controls

Forward/reverse-complement disagreement was compared with three-seed variability, the boundary heuristic $-\hat{\beta} - 0.5$, and a random control using risk-coverage curves. Analyses were repeated on the unbounded M scale and within predicted-

β strata to distinguish uncertainty signal from the mechanical compression of error near 0 and 1.

The shifted-target analysis compared fixed predictions with median methylation from 699 TCGA-BRCA primary tumours whose participants were absent from the 97 normal samples; all 791 tumour samples are reported in the Supplement. Because the sequence and MCF-10A inputs do not change, this tests degradation after tumour-associated target shifts, not generalisation to new tumour inputs.

JASPAR CORE vertebrate motifs [27] were scanned on both strands across the 42,866 GENOA pairs. Mutant scores were evaluated at the same offset and strand as the best wild-type occurrence. Per-factor Spearman correlations between motif-score change and ΔM were Benjamini–Hochberg corrected; composition, distance, substitution, and motif redundancy controls are detailed in the Supplementary Methods.

CpGenie [6] and DeepCpG [3] were reimplemented from released source and trained on the same splits, windows, augmentation, loss, and validation-selected grids. DeepCpG used its DNA module because measured neighbouring methylation is allele-invariant. Architecture and tuning details are in the Supplement.

Two pre-registered tests compared 35 held-out non-truncating ClinVar-pathogenic variants and 32 breast-cancer GWAS variants with backgrounds matched on substitution class, CpG effect, and CpG distance. Backgrounds comprised 14,996 single-base substitutions across 5,000 held-out probes. The probe was the independent unit; cohort definitions, power, and allele-resolution procedures are in the Supplementary Methods.

3 Results

3.1 Multimodal fusion outperforms single-modality models

Results were obtained on the 26,570-probe test set. The fusion model had the strongest performance across all evaluation metrics and seeds. This consistent trend suggests that DNA sequence and epigenomic context complement each other in predicting DNA methylation.

The context MLP achieved beta MAE 0.1395 ± 0.0001 and ROC-AUC 0.9187 ± 0.0001 . Sequence reduced beta MAE by 21.2% to 0.1099 ± 0.0008 and increased ROC-AUC to 0.9569 ± 0.0007 . Fusion further reduced beta MAE by 9.6% to 0.0993 ± 0.0025 and reached ROC-AUC 0.9680 ± 0.0021 (Fig. 2).

Across seeds, fusion improved sequence-only MAE by 0.01043 (95% CI 0.00951–0.01132) and ROC-AUC by 0.01053 (0.00909–0.01207).

A ridge model over 2,772 reverse-complement-collapsed k -mers matched the context model’s ROC-AUC, but fusion reduced its beta MAE by 36.6% (Table 1). Only four held-out windows had a nearest-training 31-mer Jaccard score above 0.5, and excluding MASK_snp5_common probes left fusion beta MAE unchanged while slightly raising ROC-AUC, arguing against composition, near-duplicate memorization, and

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Figure 2: Held-out chromosome performance. Gray lines connect the same training seed. (A) M-value MAE. (B) beta MAE. (C) ROC-AUC.

Table 1: Held-out chr8–9 performance against classical baselines and published methylation architectures. Mean $\pm$ SD across three seeds on 26,570 CpGs. Every model is fitted on the identical training split with hyperparameters selected on the identical validation split, so no tissue or data advantage separates them. The ridge baselines have their regularisation strength tuned; CpGenie and DeepCpG were reimplemented from their published source and tuned over the authors’ own declared grids (Section 2.4). Parameter counts: CpGenie 2.0 M, DeepCpG 4.1 M, DNABERT-2 encoder 117 M.

Metric	Composition (3 feat.)	k -mer ridge (2,772 feat.)	Context (9 feat.)	CpGenie (reimpl.)	DeepCpG (DNA module)	Sequence DNABERT-2	Fusion full
M-value MAE	1.9600	1.6866	1.4574 ± 0.0023	1.3858 ± 0.0084	1.3412 ± 0.0092	1.1941 ± 0.0070	1.0971 ± 0.0192
Beta MAE	0.1954	0.1565	0.1395 ± 0.0001	0.1281 ± 0.0011	0.1253 ± 0.0004	0.1099 ± 0.0008	0.0993 ± 0.0025
ROC-AUC	0.8748	0.9190	0.9187 ± 0.0001	0.9406 ± 0.0005	0.9437 ± 0.0009	0.9569 ± 0.0007	0.9680 ± 0.0021

genotype artefact.

CpGenie and DeepCpG exceeded the classical baselines, but fusion lowered beta MAE 20.8% below the better published architecture and increased ROC-AUC by 0.0243 (Table 1). Because DNABERT-2 is larger and pretrained, this compares complete modelling strategies rather than architecture alone.

3.2 Reference context across regulatory conditions

Fusion reduced beta MAE in every GENCODE, CpG-island, ATAC-seq, and H3K27ac stratum. Gains ranged from 0.0088 at intergenic probes to 0.0130 at UTRs and were largest at CpG shores (0.0162), in the highest ATAC-seq quartile (0.0139), and in the highest H3K27ac quartile (0.0147; Supplementary Figure 1). This concentration in regulatory-active loci supports a complementary contribution from reference context.

Against median methylation from 699 primary tumours, fixed fusion predictions worsened from beta MAE 0.0975 to 0.1180. Two-thirds of probes shifted by less than 0.05, and error rose mainly at the largest target shifts. This is a shifted-target analysis, not evaluation on tumour-specific inputs; the lack of comparable context-only degradation remains unex-

Table 2: Independent variant evaluation on GENOA. Held-out stratum, non-CpG-altering pairs, three-seed ensemble. Brackets are 95% intervals from a block bootstrap over 1-Mb blocks. Signed ρ and direction agreement are computed on genome-wide significant pairs ($p < 5 \times 10^{-8}$, $n = 4,037$); AUROC uses all 42,866 pairs.

Metric	Fusion	Sequence
Signed ρ	+0.152 [+0.117, +0.188]	+0.150 [+0.109, +0.184]
Direction agreement	0.553 [0.536, 0.570]	0.554 [0.538, 0.573]
AUROC, distance-matched	0.570 [0.554, 0.585]	0.562 [0.546, 0.578]
AUROC, unmatched	0.600 [0.586, 0.614]	0.603 [0.589, 0.617]
<i>AUROC, distance only</i>	0.595 [0.583, 0.609]	

Table 3: Variant-effect performance in two external cohorts. Held-out stratum, significant non-CpG-altering pairs, three-seed fusion ensemble. Each cohort uses its own pre-declared significance definition. Brackets are 95% block-bootstrap intervals.

	eGTEx breast tissue-matched	GENOA blood cross-tissue, cross-ancestry
<i>n</i>	418	4,037
Signed ρ	+0.246 [+0.114, +0.402]	+0.152 [+0.118, +0.186]
Direction agreement	0.596 [0.547, 0.652]	0.553 [0.536, 0.571]
Calibration slope	+1.60 [+1.13, +2.28]	+1.17 [+0.96, +1.44]
<i>Meta-analysis: $\rho = 0.178$ [0.068, 0.283], $p = 1.6 \times 10^{-3}$, $I^2 = 0\%$</i>		

plained.

3.3 External mQTL validation across cohorts

The harmonized eGTEx breast analysis contained 76,893 held-out pairs. Among 418 significant non-CpG-altering pairs, fusion reached signed $\rho = 0.246$ (0.114–0.402) and 59.6% direction agreement (Table 3). The focused set of 81 published lead mQTLs reached $\rho = 0.610$ (0.452–0.731), but is reported separately because lead variants are enriched for larger, more precisely measured effects. On that set predicted sign separated positive from negative reported slopes at AUC 0.872 (0.780–0.947), unchanged when variants inside the annotated probe span were excluded (AUC 0.877, 54 associations). Magnitude ordering was weaker than sign: absolute predicted change correlated with absolute reported slope at only $\rho = 0.217$ (–0.014–0.421).

Predicted magnitude did not, however, separate significant eGTEx leads from matched nonsignificant ones: across 35 matched sets AUROC was 0.512 (0.380–0.646, permutation $p = 0.43$). Sign is recovered where an association exists; magnitude alone does not flag which leads are significant.

Among 4,037 genome-wide significant non-CpG-altering GENOA pairs, fusion achieved signed $\rho = 0.152$ (0.117–0.188) and 55.3% direction agreement (Table 2). Agreement was weaker than in breast but remained above chance despite differences in tissue and ancestry.

Meta-analysis yielded pooled $\rho = 0.178$ (0.068–0.283, $p = 1.6 \times 10^{-3}$) without detectable heterogeneity ($I^2 = 0\%$), supporting replication across the two distinct cohorts.

Regression slopes excluded zero in both cohorts but are not calibration coefficients because the published and predicted effect scales differ. In GENOA, agreement increased with association strength and returned to chance for $p > 0.5$

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Figure 3: Agreement with GENOA rises with association strength and reaches chance where associations are null. Signed Spearman correlation (left) and direction agreement (right) across seven strata of reported p -value, with 95% block-bootstrap intervals.

($\rho = -0.004$; Fig. 3, Supplementary Table 2); it was strongest within 100 bp and reached chance near the window edge (Supplementary Table 3). The corresponding eGTEx stratum does not return to chance ($\rho = +0.019$, 0.006–0.032); the magnitude is negligible and the gradient decays smoothly rather than collapsing, as expected when 50–100 donors leave real but undetectable mQTLs above threshold, so the null-stratum claim rests on GENOA.

3.4 Distance-aware discrimination and model comparison

Distance was a major confound: it alone separated significant from null GENOA pairs at AUROC 0.595 (0.583–0.609), similar to the model’s unmatched AUROC of 0.600 (0.586–0.614). After matching every significant pair to a null pair at the same distance, fusion retained AUROC 0.570 (0.554–0.585) and remained above chance within each distance bin (Fig. 4). By contrast, the position-blind k -mer ridge fell to AUROC 0.503 (0.488–0.518), despite a signed correlation of 0.100 with measured effects. SilentMethyl therefore captures a modest positional signal beyond proximity and sequence composition; its larger unmatched AUROC should not be interpreted independently of distance.

CpGenie and DeepCpG recovered similar external signal. Only fusion versus DeepCpG on GENOA distance-matched AUROC differed significantly: +0.018 (0.004–0.032); fusion versus CpGenie was +0.004 (–0.014–0.020). Thus, stronger baseline accuracy did not translate into clearly stronger allele-specific prediction, and distance explained most marginal discrimination.

3.5 Mechanistic interpretation, uncertainty, and motif controls

Fusion substantially improved baseline methylation but not paired variant effects. Fusion-minus-sequence differences were +0.0026 (–0.0020–+0.0070) for signed ρ , –0.0007

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Figure 4: Discrimination survives distance matching. AUROC for separating genome-wide significant meQTLs from tested-but-null pairs, unmatched and within distance strata, against the distance-only baseline.

(-0.0064 – $+0.0051$) for direction agreement, and -0.0021 (-0.0053 – $+0.0011$) for distance-matched AUROC. These narrow intervals bound any improvement to less than 0.007 and support practical equivalence.

This follows from the architecture. The context vector depends on the probe, not on the allele, and is therefore identical for wild-type and mutant input. Writing the fused embedding difference,

$$\Delta \mathbf{z} = [\tilde{\mathbf{h}}_{\text{seq}}^{\text{mut}} g_{\text{seq}}^{\text{mut}} - \tilde{\mathbf{h}}_{\text{seq}}^{\text{wt}} g_{\text{seq}}^{\text{wt}}] + \tilde{\mathbf{h}}_{\text{ctx}}[g_{\text{ctx}}^{\text{mut}} - g_{\text{ctx}}^{\text{wt}}],$$

the context term survives only through a gate shift that is itself driven by the sequence change. Context can modulate a variant effect; it cannot create one.

Fusion-minus-sequence differences were also flat across gate-DNA-share quartiles. This explains their agreement among the 440 synonymous candidates ($\rho = 0.981$; 96.6% sign agreement) and shows why paired-allele fusion models with invariant covariates require a sequence-only ablation.

Forward/reverse-complement disagreement provided a single-model uncertainty signal, correlating with per-locus error at approximately $\rho = 0.50$ and retaining 75–83% of the incremental information from three-seed variability. However, a boundary heuristic, $-\lvert\hat{\beta} - 0.5\rvert$, appeared even stronger on the beta scale because absolute beta error is compressed near 0 and 1. On the unbounded M scale, the heuristic’s within-stratum correlation collapsed from 0.0369 to 0.0099, while cross-seed variability and orientation disagreement were stable (Fig. 5). We therefore evaluate uncertainty in logit space, where performance is less affected by the bounded target.

Across 831 motifs, the strongest couplings between motif-score change and predicted methylation change shared the ETS core GGAA (Supplementary Table 4; Fig. 6). Motif weakening accompanied predicted hypermethylation, consistent with loss of protective binding. Yet the k -mer ridge reproduced these couplings more strongly than the neural models (ELF4 -0.467 versus -0.266). The association therefore demonstrates sensitivity to GGAA composition, not learned binding-site grammar.

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Figure 5: Scale stability separates uncertainty signal from metric artifact. Within-stratum rank correlation between each uncertainty estimator and absolute error, on the bounded β scale and the unbounded M scale. Genuine estimators are scale-stable; the boundary heuristic is not.

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Figure 6: Motif disruption and predicted variant effects. Left: per-factor coupling between change in relative motif score and $\Delta \hat{M}$, with 95% block-bootstrap intervals; asterisks mark factors with previously reported methylation-sensitive binding. Right: meQTL discrimination among strongly versus weakly disrupting variants, which does not differ.

The two model classes differed elsewhere. In the motif-scanned cohort, the k -mer model achieved meQTL-discrimination AUROC 0.516 (0.496–0.535), compared with 0.592 for fusion. A position-blind count model can reproduce composition effects but cannot represent where a substitution occurs relative to the CpG. This separates the neural model’s positional resolution from the motif composition signal.

Disruption magnitude was unrelated to predicted effect ($\rho = -0.011$, -0.023 – $+0.002$), and strongly disrupting variants showed no greater meQTL discrimination. GC, substitution, and global-offset controls are reported in the Supplement.

The top matrices overlapped and should be read as one ETS core rather than independent factors. Previously reported methylation-sensitive factors were enriched above chance but ranked below this group.

3.6 Synonymous variant prioritization and the *NCOA2* case study

The synonymous screen uses held-out chromosomes, whereas the later *STK11* examples use training probes and illustrate only the scoring workflow.

We ranked 440 synonymous variant–CpG pairs. Cross-seed rank correlations were 0.681–0.699, while fusion and sequence-only rankings were highly concordant (signed $\rho = 0.981$, absolute $\rho = 0.966$, 96.6% sign agreement), again indicating that paired responses were driven mainly by sequence.

Median $|\Delta\hat{\beta}|$ was 0.00360 for promoter/TSS, 0.00261 for UTR, and 0.00165 for gene-body variants (Supplementary Figure 2A). Distance showed a stronger gradient: the median fell from 0.00512 within 50 bp to 0.00111 at 251–500 bp (Supplementary Figure 2B).

The highest-ranked synonymous candidate was chr8:g.70148394G>A in *NCOA2*, a GGC>GGT substitution on ENST00000452400.7. It lies in the gene body, 9 bp from OpenSea probe cg20699548, which is in the highest quartile of both ATAC-seq and H3K27ac signal. The three-seed fusion ensemble predicted a hypomethylating response of $\Delta\hat{\beta} = -0.1798 \pm 0.0198$.

Its response exceeded all 66 matched synonymous variants (Supplementary Figure 3). Because *NCOA2* depletion reduces breast-cancer cell proliferation [29], the pair motivates direct measurement of cg20699548 methylation and *NCOA2* expression, without constituting a causal claim.

3.7 Nonsynonymous extension and clinical interpretation

The same paired scoring procedure applies to nonsynonymous SNVs. In a screen of reported variants in breast-cancer-related genes, the two largest predicted responses were in *STK11*, a gene associated with hereditary breast-cancer risk in Peutz–Jeghers syndrome [30]. rs2145420809 (p.Asn119Ser) was immediately adjacent to shelf probe cg16601904 and produced $\Delta\hat{\beta} = 0.0774$; its probe was in the second ATAC-seq and third H3K27ac quartiles. rs148928808 (p.Tyr292Ter) was 204 bp from shore probe cg08681293 and produced $\Delta\hat{\beta} = -0.0613$; its probe was in the third ATAC-seq and second H3K27ac quartiles. Both variants are in the gene body on transcript NM_000455.5.

The biological relevance of rs2145420809 remains uncertain, whereas rs148928808 is a truncating variant reported in Peutz–Jeghers syndrome. In this study, however, the examples are not independent validation: both target probes belong to the training split. Their value is to demonstrate how nonsynonymous variants can be placed within the complete scored background and selected for follow-up (Fig. 7).

Predicted magnitude did not classify clinically annotated variants. None of 35 held-out non-truncating ClinVar-pathogenic variants exceeded its matched 0.05 tail ($p = 0.42$); the cohort represented only six effective probes. Three of 32 breast-cancer GWAS variants exceeded the tail ($p = 0.21$), with mean percentile 54.9 (44.8–65.2; $p = 0.36$).

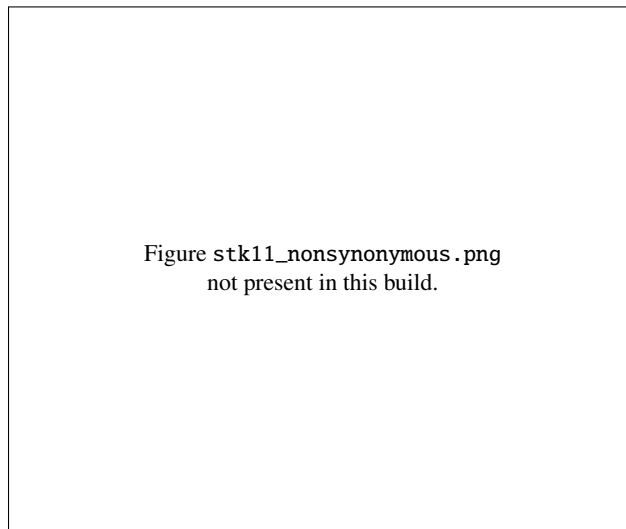


Figure 7: *STK11* variants within nonsynonymous background mutations. The histogram shows predicted WT–MUT methylation changes across SNVs in cancer-related genes. Colored lines identify rs2145420809 with cg16601904 and rs148928808 with cg08681293. Both targets are on training chromosomes and are illustrative applications rather than direct validation.

A third, better-powered test agrees: labelling every held-out mQTL by GWAS Catalog overlap and matching on variant–CpG distance and allele frequency, predicted magnitude did not separate associated from unassociated variants in either cohort (eGTEx AUROC 0.503, 0.474–0.537, 510 labelled pairs; GENOA 0.516, 0.492–0.546, 1,535 pairs). SilentMethyl therefore ranks predicted methylation consequence, not pathogenicity or disease risk.

4 Discussion

SilentMethyl establishes a unified framework for two complementary tasks in regulatory genomics: predicting baseline CpG methylation and estimating how that methylation state responds to single-nucleotide variation. Across held-out chromosomes, integrating local DNA sequence with breast epigenomic context consistently improved methylation prediction over either modality alone, demonstrating that sequence and regulatory context provide complementary information about CpG state.

At the same time, paired WT–MUT scoring enabled the same framework to move beyond locus-level prediction and prioritize individual variants according to their predicted effects on local methylation. Together, these results connect baseline epigenetic state prediction with variant-level regulatory interpretation within a single modeling framework.

Fusion produced the largest gains at CpG shores and at loci with high ATAC-seq or H3K27ac signal. Shores often exhibit tissue- and disease-associated methylation variation, while accessibility and H3K27ac mark active regulatory regions [1, 2, 31]. This concentration supports the view that reference context contributes most where sequence alone is less sufficient.

Predictions agreed modestly with measured effects across GENOA and eGTEx (pooled $\rho = 0.178$), and GENOA agreement returned to chance in the null stratum. Because distance explained much of the unmatched discrimination, the key residual result is AUROC 0.570 after exact matching. CpGenie, DeepCpG, and SilentMethyl recovered similar residual signal, while the position-blind k -mer baseline returned to chance; the evidence therefore supports positional sequence-model sensitivity, not unique superiority of SilentMethyl.

Fusion improved baseline prediction but not paired effects because context was identical for both alleles. Uncertainty was more meaningful on the unbounded M scale, and the k -mer reproduction of the ETS association limits its interpretation to GGAA composition sensitivity.

The held-out *NCOA2* candidate combines a large response, close CpG proximity, and active context into a testable hypothesis [29]. The training-split *STK11* examples extend the workflow to nonsynonymous variants but are not validation; neither case establishes causality without direct perturbation.

Several limitations define the current scope of these findings. Methylation targets were derived from cohort medians of 97 TCGA solid-tissue-normal breast samples collected adjacent to tumors and may therefore retain field effects or cell-composition heterogeneity [18, 32]. The epigenomic context was derived from MCF-10A reference tracks rather than matched individual samples, and HM450 measurements cover only a subset of genomic CpGs. Evaluation was also performed using one chromosome-holdout design, and the external mQTL benchmark represents germline variation whereas the TCGA prioritization analysis includes somatic variants. These limitations motivate validation in independent breast cohorts, broader CpG coverage, patient- or cell-type-specific epigenomic data, additional genomic and tissue settings, and ultimately direct experimental testing of highly ranked SNV-CpG pairs.

GENOA differs in tissue and ancestry, normalized effects prevent direct calibration, motifs were computationally predicted, and the ClinVar test had six effective probes. GWAS overlap is a noisy proxy for causality, since most catalog entries tag rather than cause an association. Direction agreement did not improve across quintiles of reference-effect precision (0.521 to 0.566, intervals overlapping), so the ceiling reflects model resolution rather than noise in the reported effects. Repeated chromosome partitions were not evaluated, so intervals describe one held-out split.

5 Conclusion

In summary, SilentMethyl is capable of predicting methylation along with paired differences caused by a single variant in unseen chromosome regions. Fusion outperformed sequence-only and context-only baselines for single-probe prediction, and also outperformed the CpGenie and DeepCpG architectures retrained on identical data. Predicted mutation responses from the sequence baseline and fusion model agreed with independent mQTL data in terms of direction and relative or-

dering, replicating in a tissue-matched breast cohort and in a cross-tissue, cross-ancestry cohort with no detectable heterogeneity. Finally, SilentMethyl was able to rank 440 distinct synonymous variants by prediction difference, creating hypotheses of regulatory significance to be tested experimentally, while two pre-registered tests establish that predicted magnitude alone does not identify clinically annotated variants.

6 Data and Code Availability

Code, run instructions, and Supplementary Data are available at <https://github.com/ilikecheezits/SilentMethyl>. Primary resources are cited from the GDC, eGTEx, GENOA, ENCODE, GENCODE, UCSC, and HM450 annotation [19, 21, 24, 25, 28, 31, 33]. The Supplement contains harmonised pairs, per-seed scores, evaluation outputs, and detailed analysis procedures.

References

references.bib was not available when this draft was built, so citation keys are shown in place of formatted references. Building alongside the project bibliography restores the full list. Keys marked † are new in this revision and must be added.

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